MODULE 3 UPPER LIMB MUSCLES

Muscles of the Upper Extremity.

SHOULDER, ARM, FOREARM, HAND

From the scapula to the fingertips. Learn these by compartment, not one at a time: the anterior compartment of the forearm flexes, the posterior extends, and once you know the compartment you can predict the action and the nerve of almost any muscle in it.

BY THE END

1. Name the four rotator cuff muscles and the movement each produces.
2. Identify the muscles of the arm by compartment and state the action of each.
3. Identify the muscles of the forearm by compartment and by layer.
4. Name the intrinsic muscles of the hand by group.
5. Predict the action of an unfamiliar muscle from the compartment it lies in.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: Upper limb muscles

drsrennie-stack.github.io/new-build-bio4-solano/muscular-system-concept-videos.html

The Shoulder

The rotator cuff

Four muscles run from the scapula to the head of the humerus and hold it in the shallow glenoid cavity. They are the reason the shoulder has the range it does, and the reason it dislocates more than any other joint.

THE FOUR ROTATOR CUFF MUSCLES

Muscle	Origin	Insertion	Action
Supraspinatus	Supraspinous fossa of the scapula	Greater tubercle of the humerus	Initiates abduction of the arm, the first fifteen degrees
Infraspinatus	Infraspinous fossa of the scapula	Greater tubercle of the humerus	Laterally rotates the arm
Teres minor	Lateral border of the scapula	Greater tubercle of the humerus	Laterally rotates the arm
Subscapularis	Subscapular fossa, the anterior surface of the scapula	Lesser tubercle of the humerus	Medially rotates the arm

KEY POINT

SITS: Supraspinatus, Infraspinatus, Teres minor, Subscapularis. Three insert on the greater tubercle and only subscapularis reaches the lesser tubercle, which is also the only one that rotates the arm medially. The odd insertion and the odd action are the same muscle.

OTHER MUSCLES CROSSING THE SHOULDER

Muscle	Origin	Insertion	Action
Deltoid	Lateral clavicle, acromion, spine of the scapula	Deltoid tuberosity of the humerus	Anterior fibers flex and medially rotate, middle fibers abduct, posterior fibers extend and laterally rotate
Teres major	Inferior angle of the scapula	Medial lip of the intertubercular sulcus	Extends, adducts, and medially rotates the arm
Coracobrachialis	Coracoid process of the scapula	Medial surface of the humeral shaft	Flexes and adducts the arm

The Arm

Two compartments, split by the intermuscular septa. Anterior flexes the elbow, posterior extends it, and the nerve follows the compartment.

MUSCLES OF THE ARM

Muscle	Origin	Insertion	Action
Biceps brachii	Long head from the supraglenoid tubercle, short head from the coracoid process	Radial tuberosity and the bicipital aponeurosis	Flexes the elbow and supinates the forearm; also flexes the arm
Brachialis	Distal anterior humerus	Coronoid process and tuberosity of the ulna	Flexes the elbow, the prime mover regardless of forearm position
Triceps brachii	Long head from the infraglenoid tubercle, lateral and medial heads from the humerus	Olecranon of the ulna	Extends the elbow; the long head also extends the arm
Anconeus	Lateral epicondyle of the humerus	Lateral olecranon and posterior ulna	Assists extension of the elbow
Brachioradialis	Lateral supracondylar ridge of the humerus	Styloid process of the radius	Flexes the elbow, strongest with the forearm in mid-position

KEY POINT

Brachioradialis flexes the elbow but sits in the posterior compartment and is supplied by the radial nerve. It is the exception that proves the compartment rule, and it is asked about for exactly that reason.

The Forearm

Anterior compartment, the flexors and pronators

ANTERIOR FOREARM, SUPERFICIAL TO DEEP

Muscle	Origin	Insertion	Action
Pronator teres	Medial epicondyle of the humerus and coronoid process of the ulna	Lateral mid-shaft of the radius	Pronates the forearm and assists elbow flexion
Flexor carpi radialis	Medial epicondyle	Bases of metacarpals 2 and 3	Flexes and abducts the wrist
Palmaris longus	Medial epicondyle	Palmar aponeurosis	Tenses the palmar aponeurosis; absent in many people
Flexor carpi ulnaris	Medial epicondyle and olecranon	Pisiform, hamate, metacarpal 5	Flexes and adducts the wrist
Flexor digitorum superficialis	Medial epicondyle, coronoid process, radius	Middle phalanges of digits 2 to 5	Flexes the proximal interphalangeal joints
Flexor digitorum profundus	Anterior and medial ulna and interosseous membrane	Distal phalanges of digits 2 to 5	Flexes the distal interphalangeal joints
Flexor pollicis longus	Anterior radius and interosseous membrane	Distal phalanx of the thumb	Flexes the thumb
Pronator quadratus	Distal anterior ulna	Distal anterior radius	Pronates the forearm, the prime mover of pronation

Posterior compartment, the extensors and the supinator

POSTERIOR FOREARM

Muscle	Origin	Insertion	Action
Extensor carpi radialis longus	Lateral supracondylar ridge	Base of metacarpal 2	Extends and abducts the wrist
Extensor carpi radialis brevis	Lateral epicondyle	Base of metacarpal 3	Extends and abducts the wrist
Extensor digitorum	Lateral epicondyle	Extensor expansions of digits 2 to 5	Extends the fingers and the wrist
Extensor carpi ulnaris	Lateral epicondyle and posterior ulna	Base of metacarpal 5	Extends and adducts the wrist
Supinator	Lateral epicondyle and proximal ulna	Proximal lateral radius	Supinates the forearm
Abductor pollicis longus	Posterior radius, ulna, interosseous membrane	Base of metacarpal 1	Abducts and extends the thumb
Extensor pollicis brevis	Posterior radius and interosseous membrane	Proximal phalanx of the thumb	Extends the thumb at the metacarpophalangeal joint
Extensor pollicis longus	Posterior ulna and interosseous membrane	Distal phalanx of the thumb	Extends the thumb at the interphalangeal joint
Extensor indicis	Posterior ulna and interosseous membrane	Extensor expansion of digit 2	Extends the index finger

KEY POINT

Almost every superficial flexor takes origin from the medial epicondyle, and almost every superficial extensor from the lateral epicondyle. Learn the two common origins first and the individual muscles become variations rather than a list of twenty.

The Hand

The intrinsic muscles lie entirely within the hand. Learn them as groups.

THE INTRINSIC MUSCLES, BY GROUP

Group	Muscles	Action
Thenar	Abductor pollicis brevis, flexor pollicis brevis, opponens pollicis	Move the thumb; the bulge at the base of the thumb
Hypothenar	Abductor digiti minimi, flexor digiti minimi brevis, opponens digiti minimi	Move the little finger; the bulge on the ulnar side of the palm
Midpalmar	Lumbricals, palmar interossei, dorsal interossei	Lumbricals flex the knuckles and extend the fingers; palmar interossei adduct, dorsal interossei abduct
Adductor pollicis	A single muscle in the web space	Adducts the thumb

KEY POINT

PAD and DAB: Palmar interossei ADduct, Dorsal interossei ABduct. The reference line for abduction and adduction in the hand is the middle finger, not the midline of the body.

COMPARTMENT, NERVE, AND THE PATTERN OF LOSS

The **supraspinatus** tendon passes under the acromion, a bony roof with no give. Repeated overhead work pinches it there, which is why impingement and cuff tears concentrate in that one tendon of the four.

Tennis elbow is inflammation at the **lateral epicondyle**, the common origin of the wrist extensors. Golfer's elbow is the same problem at the **medial epicondyle**, the common origin of the wrist flexors. Two names, one anatomical idea.

The **flexor digitorum profundus** reaches the distal phalanx and the **flexor digitorum superficialis** stops at the middle one. That is how a clinician tests them separately, by holding the other fingers straight and asking the patient to bend one joint at a time.

The **thenar** group wastes visibly in long-standing carpal tunnel syndrome, because the median nerve supplies it and the nerve is compressed at the wrist. A flat thenar eminence is a structural sign of a nerve problem.